

DECODED EXPLAINED

5th September, 2026

India and Belgium Ramp Up Bilateral Defence Cooperation



MEGHNA BAL

META RECENTLY settled a \$17.1 billion case with several US states concerning allegations about its impact on children's mental health. The company must make several changes to its apps, including incorporating daily usage limits and "nighttime blocks" for teenagers, "enhanced age assurance measures" to prevent children from using them and the de-personalisation of feeds, meaning that a child's feed would serve random content rather than content tailored to their interests.

Many have hailed the decision as a step towards holding platforms accountable. But such framing is problematic because it places the platform, rather than the child, at

Focus on the children, not the platform

the centre of the problem. Once platforms become the focus, the policy response gravitates towards imposing more obligations on them. Asking what more platforms can be required to do is not the same as asking what children really need to be safer online.

The first problem with a platform-centric child safety approach is that it often produces measures that are difficult to enforce. The requirement for Meta to introduce age-gating is one example. Children are sophisticated and motivated users of technology and find ways around such measures.

The second is that such an approach can ignore what children need to be safe or have positive experiences online. The move to restrict personalisation illustrates this tension. Poorly designed recommendation systems can expose children to harmful content, but removing personalisation does not auto-

matically make a service safer. It may expose children to irrelevant or inappropriate material, while making it harder for educational or support services to reach them.

Instead of focusing on eliminating every risk from the online environment, decision-makers must look at how to equip children to deal with those risks as we do in the physical world: What not to share, how to respond to unsolicited contact, how to recognise ma-

For many children, digital spaces provide support that may not be available offline. LGBTQIA+ children and those experiencing abuse may use online platforms to find information or help

nipulation, when to block or report someone, and when to seek help. Yet digital literacy and sensitisation receive far less policy attention. India has made a nod to such an approach, with the 2026 Economic Survey calling for a digital wellness curriculum in schools. It remains to be seen how widely and appropriately this will be implemented.

The third problem is that for many children, digital spaces provide community, expression and support that may not be available offline. LGBTQIA+ children, those experiencing neglect or abuse, and socially isolated children may use online platforms to find information, companionship or help. Children's online participation is also not confined to one platform. If one service becomes too restrictive or cumbersome, they can simply migrate to platforms with weaker protections.

The fourth problem is that children's online engagement is often shaped by their offline environment. In India, screens may not necessarily be displacing abundant offline opportunities but filling a vacuum. Many children lack access to safe public spaces or affordable recreational facilities. If policy-makers want children to spend less time online, they must also ask what children are being offered offline.

None of this absolves platforms of responsibility. But approaches more in touch with children's engagement online are likely to be more effective. This requires addressing the entire ecology of childhood: Platforms, parents, schools, communities, public spaces and children themselves.

The writer is director of the Esya Centre, a New Delhi-based think tank. Views are personal

Focus on the children not the platform

- **Why focussing on platforms is not enough ?**
 - **Difficult to enforce**
 - **It could ignore the safety net developments for the online experience of children**
 - **Sometimes digital spaces provide space for community expression and support that may be not available offline**
 - **Often screen time is result of lack of offline opportunities and focussing on platforms are not enough**
- **Way ahead**
 - **Digital Media Literacy Education : Developing the ability to question whether content is accurate, realistic or fictional is therefore essential.**
 - **Parents have an important role to play**
 - **Modeling healthy behavior is key : Regardless of age, children learn more from what they see than from what they are told.**
 - **Safe-by-design approach to social media**

Indian researchers map colonial spider venom, find anti-cancer potential

Indian researchers map colonial spider venom, find anti-cancer potential

Sarath Babu George
THIRUVANANTHAPURAM

A team of Indian researchers has produced the first comprehensive molecular and functional profile of the venom gland of *Stegodyphus sarasinorum*, a colonial spider species found across the Indian subcontinent, to identify numerous proteins and small molecules that could serve as promising leads for future drug discovery.

Published in Springer Nature's peer-reviewed, open-access journal *Scientific Reports*, the study identifies bioactive compounds associated with anti-cancer, antimicrobial and immune-related func-



Stegodyphus sarasinorum

tions, while also demonstrating that venom gland extracts can kill cancer cells under laboratory conditions.

The researchers caution that the findings require extensive validation before any clinical application.

Lead researcher A.P.

Ajay Kumar of the Department of Zoology at Sree Neelakanta Government Sanskrit College, Pattambi, Keralam, led the multidisciplinary study.

S. sarasinorum lives in permanent colonies, constructs communal silk nests and captures prey cooperatively. It is among a small number of permanently social spider species known worldwide, thereby making it an important model for studying spider behaviour and venom biology.

The analysis identified 31 protein components through transcriptomic profiling and 32 proteins through proteomic analysis. In addition Ultra High-

Performance Liquid Chromatography-Mass Spectrometry (UHPLC-MS) detected 81 metabolites, including amino acids, biogenic amines and organic acids to provide a detailed molecular inventory of the venom gland.

Cytotoxicity

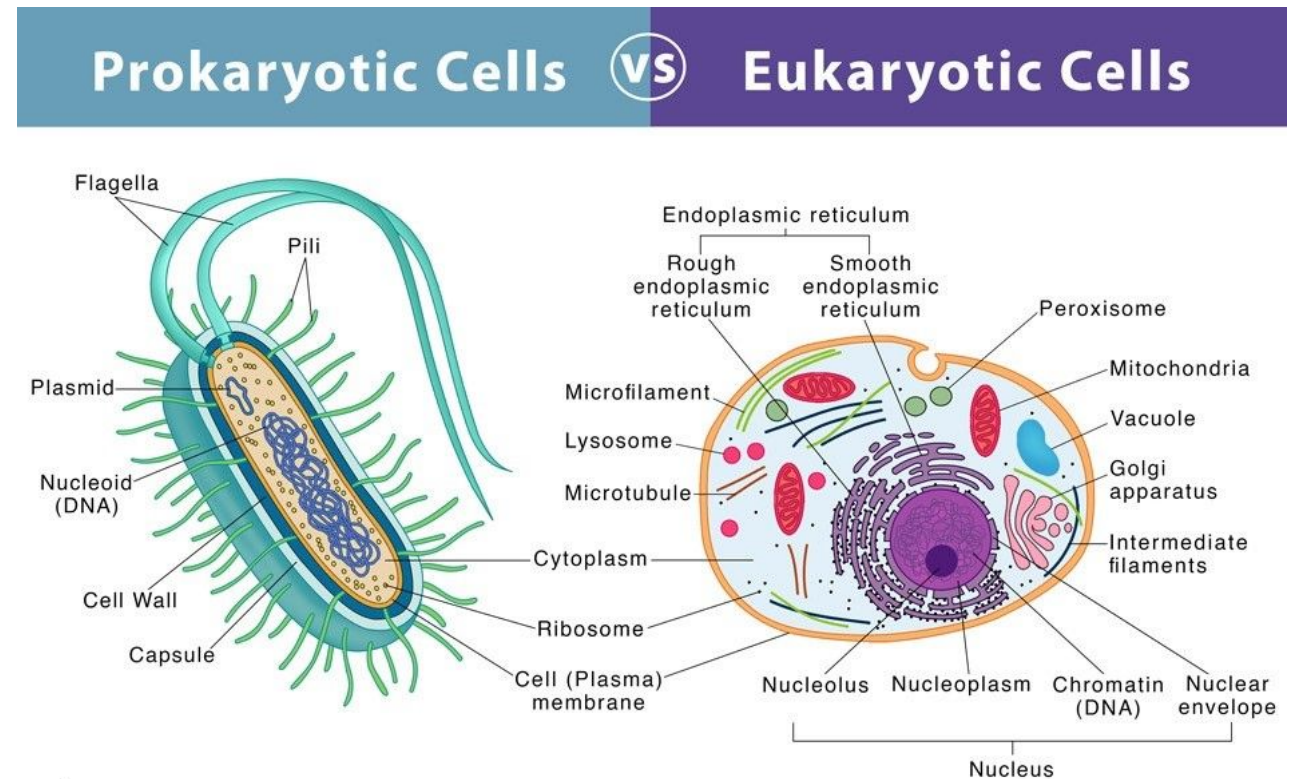
To investigate the biological activity of these molecules, the researchers tested venom gland extracts against Dalton's Lymphoma Ascites (DLA) cancer cells in laboratory experiments. The extracts exhibited dose-dependent cytotoxicity, meaning that higher concentrations destroyed a greater proportion of cancer cells.

Indian researchers map colonial spider venom, find anti-cancer potential

- **"Dumb Kids Prefer Candy Over Fresh Green Salad"**
 - **Dumb – Domain**
 - **Kids – Kingdom**
 - **Prefer – Phylum**
 - **Candy – Class**
 - **Over – Order**
 - **Fresh (or Fancy) – Family**
 - **Green – Genus**
 - **Salad – Species**

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- Prokaryotes are organisms that lack a true nucleus and membrane-bound organelles.
- Eukaryotes are organisms that have cells containing a well-defined nucleus enclosed within a membrane.



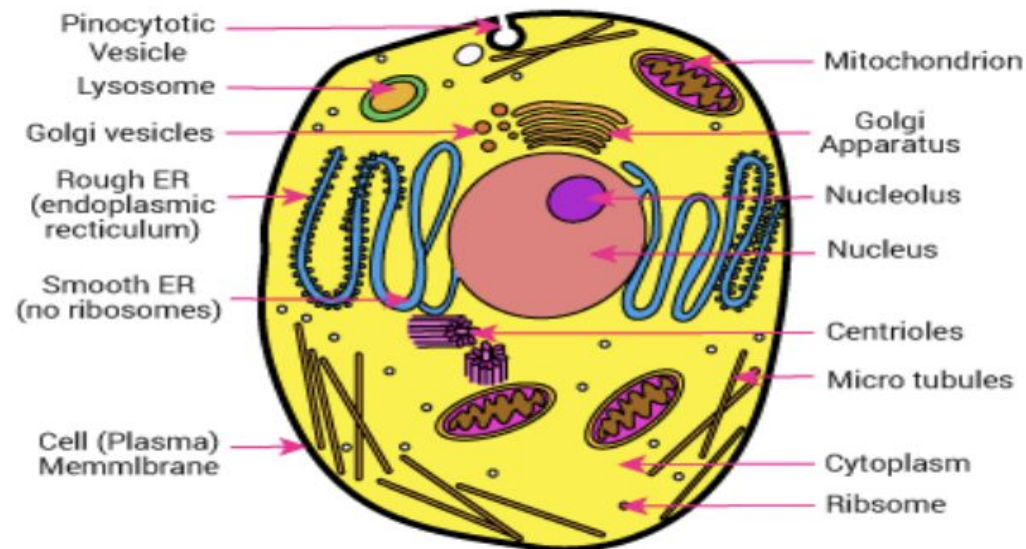
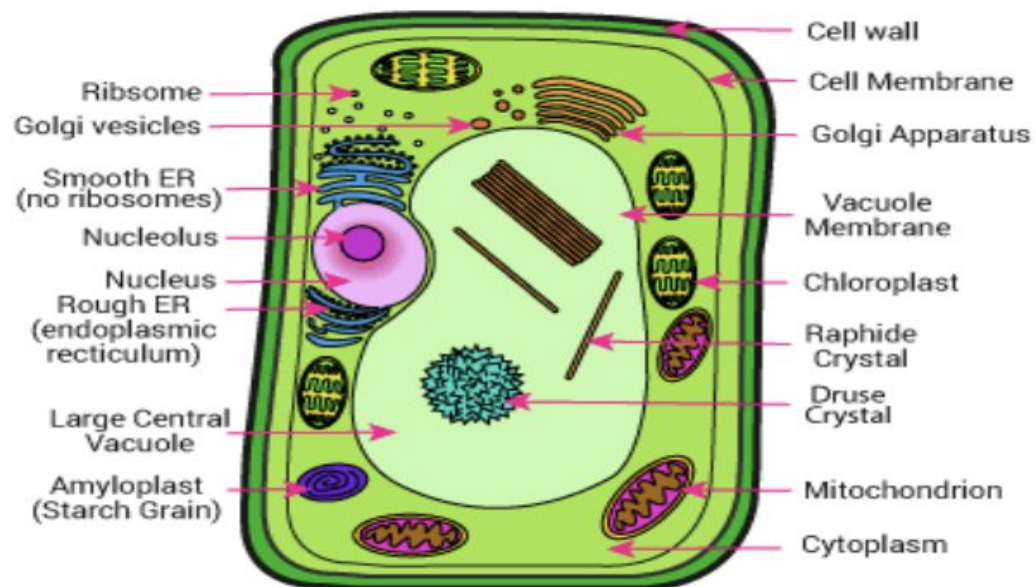
Indian researchers map colonial spider venom, find anti-cancer potential

- **Domains : Bacteria, Archaea and Eukarya**

Features	Bacteria	Archaea	Eukaryotes
Cell Type	Prokaryotes	Prokaryotes	Eukaryotes
Nucleus	No	No	Yes
Cell Wall	Present with peptidoglycan	Present without peptidoglycan	May or may not be present

Indian researchers map colonial spider venom, find anti-cancer potential

- **Kingdoms : Plantae, Animalia, Fungi, Protists and Monera**



- **Differences between Animal Cell and Plant Cell**

- While Plant cell have chloroplast Animal Cell do not have chloroplast
- While Animal cell have one or small vacuoles, Plant cells have a large central vacuole
- While Animal Cell lack Cell wall, plants have a Cell wall

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Mains Phylums under Animalia

- **Porifera** includes animals such as Sycon, Spongilla, Euplectella and Euspongia, commonly known as **sponges**, which have numerous pores on their bodies.
- **Cnidaria (Coelenterata)** includes **Hydra, jellyfish, sea anemones, corals and Obelia**, which are aquatic animals
- **Annelida** includes **earthworms, leeches**
- **Arthropoda** includes **insects such as cockroaches and butterflies, arachnids such as spiders and scorpions and tarantulas**, and **crustaceans such as crabs, prawns and lobsters**, and is characterised by jointed appendages and an external skeleton.
- **Mollusca** includes **snails, oysters, mussels, octopuses, squids and cuttlefish**, which generally have soft, unsegmented bodies.
- **Echinodermata** includes **starfish, sea urchins, sea cucumbers and brittle stars**, which are exclusively marine animals generally characterised by a spiny skin and a water vascular system.
- **Chordata** includes **fish, amphibians, reptiles, birds and mammals including humans**, and is characterised by the presence of a notochord at least during some stage of development.

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Which of the following statements are correct regarding the general difference between plant and animal cells?

1. Plant cells have cellulose cell walls while animal cells do not.
2. Plant cells do not have a plasma membrane, unlike animal cells which do.
3. Mature plant cell has one large vacuole whilst animal cell has many small vacuoles.

Select the correct answer using the code given below:

This question was previously asked in

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1. 1 and 2 only

2. 2 and 3 only

3. 1 and 3 only

4. 1, 2 and 3

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- **Social Spider - Unlike most spiders, which are solitary, *S. sarasinorum* lives in large, permanent colonies and cooperatively captures and feeds on prey.**
- **Indian Cooperative Spider - It is also known as the Indian cooperative spider.**
- **Family – Eresidae, commonly associated with velvet/social spiders.**
- **Distribution - Its recorded distribution includes Pakistan, India, Sri Lanka, Nepal and Myanmar.**
- **Habitat - It occurs particularly in dry, shrubby and seasonal habitats, with communal silk nests often constructed on shrubs and trees.**
- **Indian researchers have produced the first comprehensive molecular and functional profile of the venom gland of the colonial spider *Stegodyphus sarasinorum*, indicating anti-cancer drug potential.**

Indian researchers map colonial spider venom, find anti-cancer potential



Regarding Peacock tarantula (Gooty tarantula), consider the following statements:

- I. It is an omnivorous crustacean.
- II. Its natural habitat in India is only limited to some forest areas.
- III. In its natural habitat, it is an arboreal species.

Which of the statements given above is/are correct?

A I only

B I and III

C II only

D II and III

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- **Peacock Tarantula is a rare arboreal spider endemic to India, known for its bright blue color, unique patterns, ecological role and critically endangered status.**
- **This species is endemic to India, primarily found in parts of Andhra Pradesh.**
- **Its natural habitat in India is only limited to some forest areas, which makes the species highly vulnerable to environmental changes and habitat destruction.**
- **In its natural habitat, it is an arboreal species, meaning it lives on trees, occupying tree holes, bark crevices, and elevated surfaces.**
- **The species is listed as Critically Endangered by the International Union for Conservation of Nature (IUCN) due to its declining population.**

ISRO Launches advanced imaging satellite EOS-05

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India's first dedicated imaging satellite will provide real-time imagery from about 36,000 km above earth; ISRO breathes easy after successful mission in the wake of back-to-back failures

Hemanth C.S.
BENGALURU

The Indian Space Research Organisation (ISRO) on Friday successfully launched its advanced earth observation satellite EOS-05 aboard the Geosynchronous Satellite Launch Vehicle (GSLV-F17).

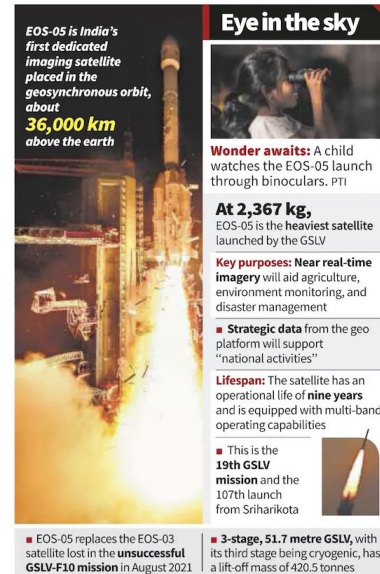
The GSLV, on its 19th flight, placed the 2,367-kg EOS-05 into the intended sub-geosynchronous transfer orbit (Sub-GTO) about 18 minutes after lifting off from the second launch pad of the Satish Dhawan Space Centre in Sriharikota at 2.55 a.m.

The EOS-05 is India's first imaging satellite designed to monitor earth from geosynchronous orbit, about 36,000 km above earth.

"I am very happy to announce that the GSLV-F17 vehicle has successfully and precisely injected the advanced earth observation satellite EOS-05 into the intended and required orbit," ISRO Chairman V. Narayanan said.

He said this was the 107th launch from the spaceport in Sriharikota and the 19th mission of the GSLV.

"The first launch of the GSLV-D1 was done with a payload of 1,536 kg. From that, we have gradually im-



proved the performance of this vehicle by optimising the structural mass and improving the propulsion systems. Today, we have injected the heaviest satellite, a 2,367-kg satellite, using this launch vehicle," he added.

Dr. Narayanan said that once operationalised, EOS-05 would become In-

dia's first dedicated imaging satellite operating from the geosynchronous orbit, providing important strategic data for the country.

'Operating perfectly'

"From the satellite data, we confirm that all the valves are operating perfectly, the solar panel is deployed, and the health of

the satellite is perfectly all right. In the coming days, we will raise the orbit and take it to the required orbit and place the satellite in the geo platform for national activities," Dr. Narayanan said.

The EOS-05, dubbed an "eye in the sky", will provide real-time imagery, helping sectors such as agriculture, environment and disaster management.

A huge relief

The successful launch comes as a huge relief for ISRO, which has not carried out a launch since the failure of the PSLV-C62 mission carrying the EOS-N1 earth observation satellite on January 12, 2026.

The space agency, following the back-to-back failures of the PSLV-C62 in January 2026 and the PSLV-C61 on May 18, 2025, had taken a cautious approach and refrained from carrying out further launches.

Though seven missions, including the GSLV-F17/EOS-05 mission, had been scheduled to take place in the first quarter of 2026, ISRO did not launch any satellites during the period.

The successful launch of the 2367-kg EOS-05 also puts behind the failure ISRO had encountered in August 2021 with the GSLV-F10/EOS-03 mission.

ISRO Launches advanced imaging satellite EOS-05

- **EOS-05 (formerly GISAT-1A) is an ISRO geo-imaging satellite built on the I-2K platform. Operating in Geosynchronous Orbit (~35,786 km), it enables near-real-time monitoring of India, surrounding oceans and border regions, succeeding the lost GISAT-1 mission.**
- **Placed into a high-altitude Geosynchronous Orbit (~35,786 km), matching Earth's rotation to overcome the multi-day revisit gaps typical of Low Earth Orbit (LEO) satellites.**
- **Functions as an Eye in the Sky by maintaining uninterrupted coverage over the Line of Actual Control (LAC), the Line of Control (LoC), and key maritime passages across the northern Indian Ocean.**

NCPCR campaign rescues over 3800 from child labour

NCPCR campaign rescues over 3,800 from child labour

The Hindu Bureau

NEW DELHI

The National Commission for Protection of Child Rights (NCPCR) has rescued over 3,800 children from child labour under a sustained campaign launched in June. Over 575 FIRs have also been registered in connection with these incidents.

The NCPCR, a statutory body under the Ministry of Women and Child Development, launched a special “Pan-India Rescue and Rehabilitation” campaign on June 12, which continued until August 31, an official statement said here on Friday.

The Child Rights Commission undertook coordinated measures with the concerned State Governments/Union Territory administrations, district magistrates, labour departments, police authorities and other stakeholders to intensify the identi-

fication, rescue and rehabilitation of children engaged in child labour and those vulnerable to trafficking and exploitation.

Following this, it issued communications to the concerned authorities requesting intensified and targeted rescue operations at child labour and trafficking hotspots.

States and UTs were also requested to furnish regular data on rescue operations, children rescued, FIRs registered, rehabilitation, restoration, educational mainstreaming, compensation and other follow-up measures.

To monitor implementation and address field-level issues, the NCPCR held meetings with the district administrations. The meetings focused on reviewing rescue operations, addressing operational bottlenecks and strengthening coordination among the concerned departments.

NCPCR campaign rescues over 3800 from child labour

- The National Commission for Protection of Child Rights (NCPCR) has been constituted by the Government of India under the Commission for Protection of Child Rights (CPCR) Act, 2005, and is mandated to function for the protection and promotion of child rights.
- The Commission is further mandated to monitor the proper and effective implementation of
 - Protection of Children from Sexual Offences (POCSO) Act, 2012
 - Juvenile Justice (Care and Protection of Children) Act, 2015
 - Right to Free and Compulsory Education (RTE) Act, 2009.
- India has acceded to the Convention on the Rights of the Child (CRC) in 1992 which is an international treaty that makes it incumbent upon the signatory States to take all necessary steps to protect children's rights enumerated in the Convention.
- **Composition**
 - Chairperson : Person of eminence and has done outstanding work for promoting the welfare of children.
 - Terms : 3 years or till the age of 65 years.
 - Provided not more than 2 terms.
 - 6 Members
 - At least two of the members should be women
 - Person of eminence in the following fields: Education, Child health, care, welfare or child development, Juvenile justice or care of neglected or marginalized children or children with disabilities, Elimination of child labour or children in distress, Child psychology or sociology and Laws relating to children
- The Chairperson and Members of the Commission are appointed by the Central government.

NCPCR campaign rescues over 3800 from child labour

- **Function and Responsibilities**
 - Examine and review existing safeguards for the protection of child rights and recommend measures for their effective implementation.
 - Report annually and at other intervals to the central government on the effectiveness of these safeguards.
 - Investigate violations of child rights and recommend legal proceedings in appropriate cases.
 - Review existing policies, programs, and activities related to child rights and make recommendations for their improvement.
 - Promote research in the field of child rights.
 - Raise awareness of child rights and available safeguards through various means, such as publications, media, and seminars.
 - Inspect institutions where children are detained or reside, including juvenile homes, and recommend remedial action if necessary.
 - Investigate complaints and take suo motu notice of issues related to the deprivation and violation of child rights and the non-implementation of laws protecting and developing children.
 - The Commission, while investigating, shall have all the powers of a Civil court in respect of the following matters
- **Recommendations**
 - After the inquiry, the Commission has the power to can take actions like
 - Recommending to the concerned Government or authority the initiation of proceedings for prosecution against the concerned person(s).
 - Approaching the Supreme Court or the High Court concerned for directions, orders, or writs.
 - Recommending to the concerned Government or authority for the grant of such interim relief to the victim or the members of the family.
- **Matters pending before state commission can be taken**

The Gap in Manufacturing Sector GVA

2025 Mains Question

The National Commission for Protection of Child Rights (NCPCR) plays a crucial role in safeguarding child rights in India. Critically assess its mandate, functioning, and limitations in ensuring justice and welfare for children. (250 words, 15 marks)

Thank You